IGCSE 0607 Practice Questions — E6.5 — Combined Questions & Markscheme

Compound shapes and parts of shapes

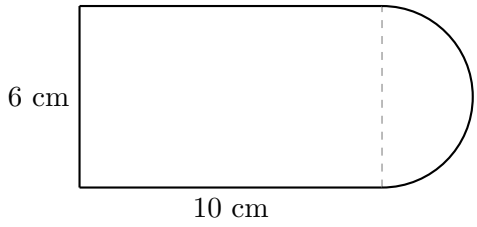
Total Questions: 36 — **Total Marks:** 267 Marks

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- **E6.5** Compound shapes and parts of shapes
  → **Paper 2 (Non-Calculator)** *(13 Qs)* · **Paper 4 (Calculator)** *(23 Qs)*

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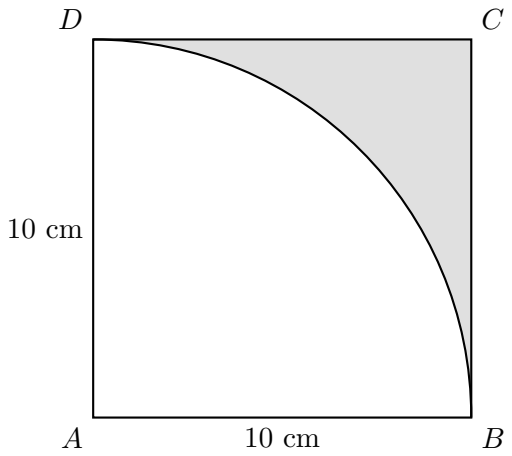
1 A shape is made from a rectangle and a semicircle.
The rectangle has a length of 10 cm and a width of 6 cm.
The semicircle is attached to one of the 6 cm sides.



Find the exact perimeter of the shape.
Give your answer in the form $a + b\pi$.

..... cm **[3]**

2 The diagram shows a square $ABCD$ with side length 10 cm.
A quarter circle with center A and radius 10 cm is drawn inside the square.
The region inside the square but outside the quarter circle is shaded.



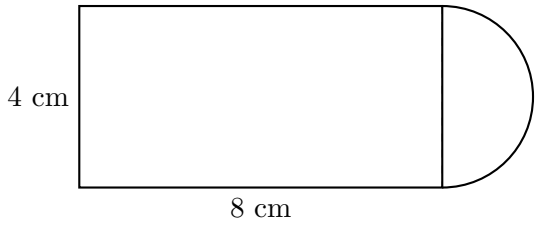
(a) Find the exact perimeter of the shaded region.
Give your answer in terms of π.

..... cm **[2]**

(b) Find the exact area of the shaded region.
Give your answer in terms of π.

..... cm^2 **[2]**

3 The diagram shows a compound shape formed by joining a semi-circle to a rectangle.
The rectangle has a length of 8 cm and a width of 4 cm.
The diameter of the semi-circle is 4 cm.



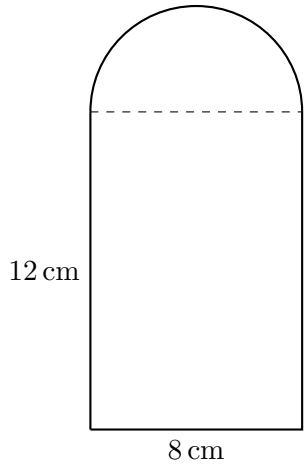
(a) Calculate the perimeter of the shape.
Give your answer in its simplest form, in terms of π.

..... *cm* **[2]**

(b) Calculate the area of the shape.
Give your answer in its simplest form, in terms of π.

..... cm^2 **[2]**

4 A compound shape is made from a rectangle and a semicircle.
The rectangle has width 8 cm and length 12 cm. The semicircle is attached to one of the sides of width 8 cm.



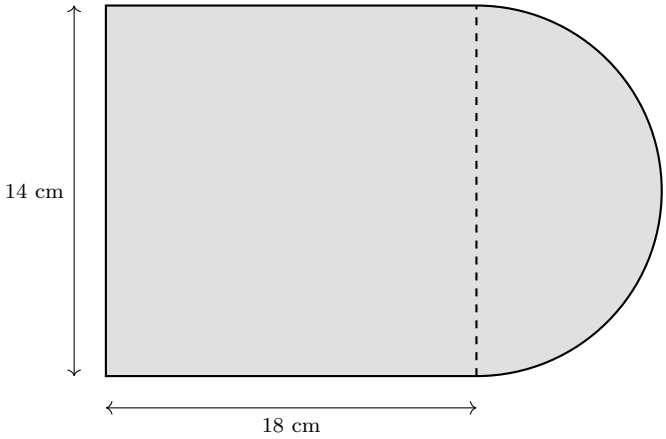
(a) Calculate the exact perimeter of the compound shape.
Give your answer in its simplest form, in terms of π.

..... cm **[2]**

(b) Calculate the exact area of the compound shape.
Give your answer in its simplest form, in terms of π.

..... cm^2 **[2]**

5 The diagram shows a compound shape made by attaching a semicircle of diameter 14 cm to one end of a rectangle of length 18 cm and width 14 cm.



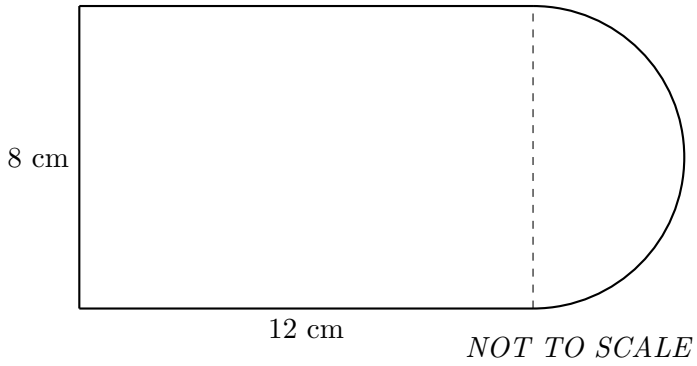
(a) Calculate the perimeter of the compound shape.
Give your answer in terms of π.

Perimeter = ... cm **[2]**

(b) Calculate the total area of the compound shape.
Give your answer in terms of π.

Area = ... cm^2 **[3]**

6 The diagram shows a shape made from a rectangle and a semicircle.
The rectangle has length 12 cm and width 8 cm.
The semicircle has a diameter of 8 cm.



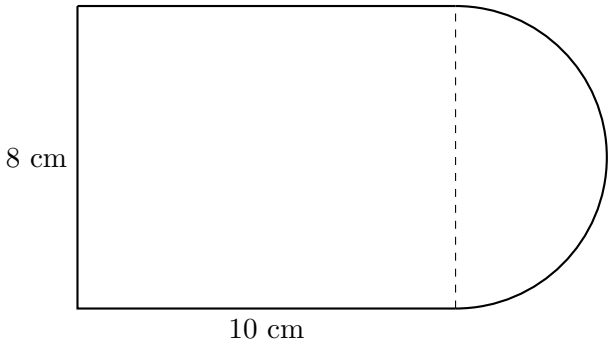
(a) Find the perimeter of the shape.
Give your answer in its simplest form in terms of π.

Perimeter = ... cm **[2]**

(b) Find the area of the shape.
Give your answer in its simplest form in terms of π.

Area = ... cm^2 **[3]**

7 The diagram shows a compound shape made from a rectangle and a semicircle.
The rectangle has a length of 10 cm and a width of 8 cm.
The semicircle is attached to one of the 8 cm sides.



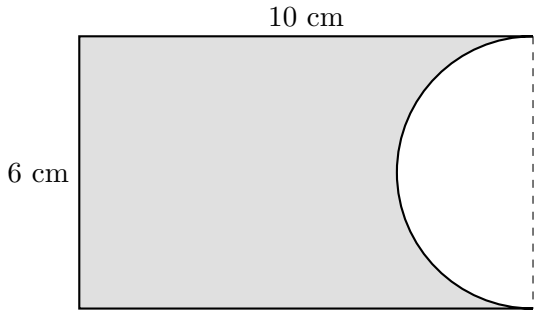
(a) Work out the exact area of the shape.
Give your answer in the form $a + b\pi$, where a and b are integers.

Area = ... cm^2 **[3]**

(b) Work out the exact perimeter of the shape.
Give your answer in the form $c + d\pi$, where c and d are integers.

Perimeter = ... cm **[3]**

8 The diagram shows a compound shape made from a rectangle and a semicircle.
The rectangle has a length of 10 cm and a width of 6 cm.
A semicircle of diameter 6 cm has been removed from one end of the rectangle.



NOT TO SCALE

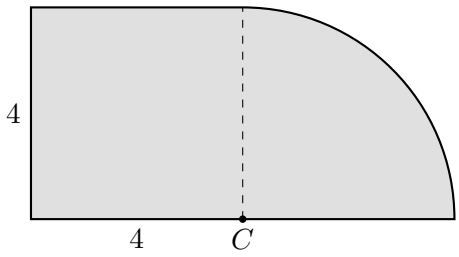
(a) Work out the perimeter of the shape.
Give your answer in the form $a + b\pi$, where a and b are integers.

..... cm **[3]**

(b) Work out the area of the shape.
Give your answer in the form $c - d\pi$, where c is an integer and d is a decimal.

..... cm^2 **[3]**

9 The diagram shows a shape formed by a square of side 4 cm and a quarter of a circle with center C.



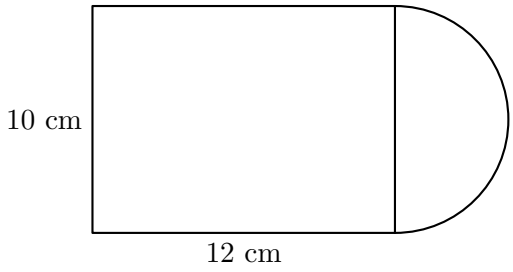
(a) Find the area of the shape.
Give your answer in terms of π.

Area = ... cm^2 **[3]**

(b) Find the perimeter of the shape.
Give your answer in terms of π.

Perimeter = ... cm **[3]**

10 The diagram shows a compound shape made from a rectangle and a semicircle.
The rectangle has length 12 cm and width 10 cm.
The semicircle has a diameter of 10 cm.



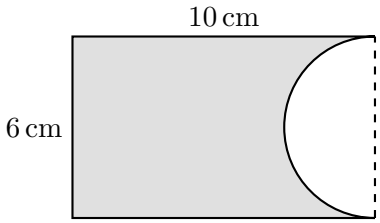
(a) Calculate the perimeter of the shape.
Give your answer in the form $a + b\pi$.

..... cm **[3]**

(b) Calculate the area of the shape.
Give your answer in the form $c + d\pi$.

..... cm^2 **[3]**

11 The diagram shows a shape made from a rectangle with dimensions 10 cm by 6 cm. A semicircle of diameter 6 cm has been removed from one end.



NOT TO SCALE

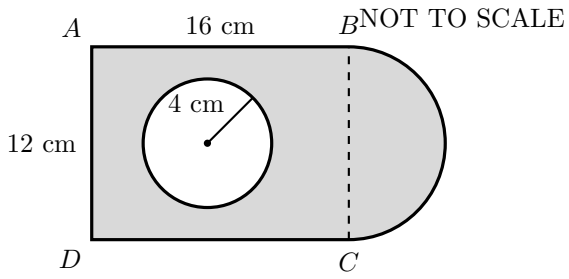
(a) Calculate the area of the shape.
Give your answer in terms of π.

..... cm^2 **[3]**

(b) Calculate the perimeter of the shape.
Give your answer in terms of π.

..... cm **[3]**

12 The diagram shows a shaded metal plate formed from a rectangle $ABCD$ with a semicircle of diameter BC attached to side BC.
Inside the rectangle, a circular hole of radius 4 cm is cut out.
$AB = 16$ cm and $BC = 12$ cm.



(a) Giving your answers in terms of π, calculate

(i) the outer perimeter of the metal plate,

..... cm **[2]**

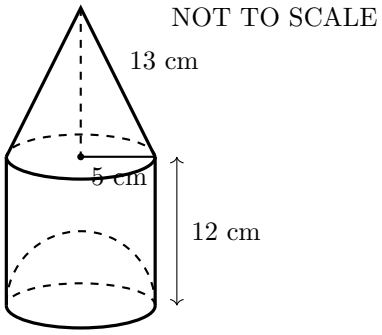
(ii) the area of the shaded region.

..... cm^2 **[3]**

(b) The shaded metal plate is the uniform cross-section of a solid prism of thickness 5 cm. Calculate the volume of the prism, giving your answer in the form $p + q\pi$, where p and q are integers.

..... cm^3 **[2]**

13 The diagram shows a compound solid formed from a right circular cylinder of base radius 5 cm and height 12 cm. A right circular cone of base radius 5 cm and slant height 13 cm is joined to the top face of the cylinder, and a hemisphere of radius 5 cm is hollowed out of the bottom face of the cylinder.



(a) Show that the vertical height of the conical part is 12 cm.

[2]

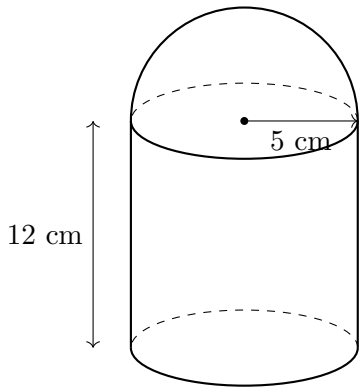
(b) Calculate the total surface area of the compound solid, giving your answer in terms of π.

..... cm^2 **[4]**

(c) Calculate the volume of the compound solid, giving your answer as a single fraction in terms of π.

..... cm^3 **[4]**

14 A solid is made by attaching a hemisphere of radius 5 cm to one end of a solid cylinder.
The cylinder has a radius of 5 cm and a height of 12 cm.



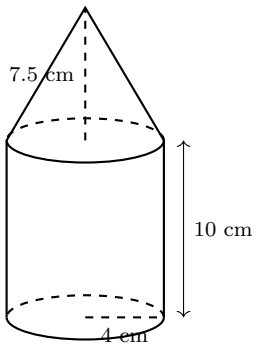
(a) Calculate the total volume of the solid.
Give your answer correct to 3 significant figures.

..... cm^3 **[3]**

(b) Calculate the total surface area of the solid.
Give your answer correct to 3 significant figures.

..... cm^2 **[3]**

15 A solid metal trophy consists of a right circular cone placed on top of a cylinder, both of base radius 4 cm.
The cylinder has height 10 cm and the cone has perpendicular height 7.5 cm.



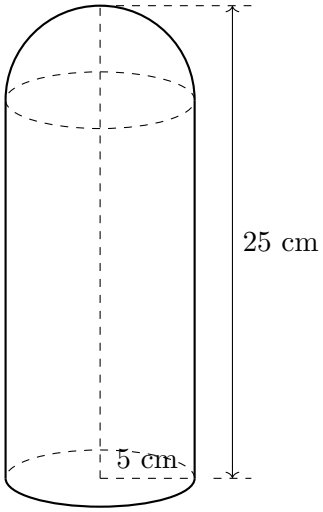
(a) Calculate the total volume of the solid trophy.

Volume = ... cm^3 **[3]**

(b) Calculate the total surface area of the solid trophy.

Surface area = ... cm^2 **[3]**

16 The diagram shows a solid in the shape of a cylinder with a hemisphere attached to the top.
The radius of both the cylinder and the hemisphere is 5 cm.
The total height of the solid is 25 cm.



NOT TO SCALE

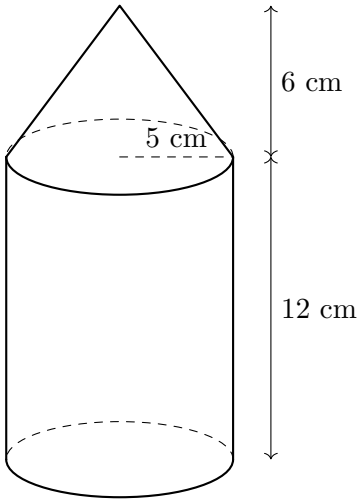
(a) Calculate the total volume of the solid.

..... cm^3 **[3]**

(b) Calculate the total surface area of the solid.

..... cm^2 **[4]**

17 The diagram shows a solid compound shape made from a cylinder and a cone.
The cylinder and the cone both have a base radius of 5 cm.
The height of the cylinder is 12 cm and the height of the cone is 6 cm.



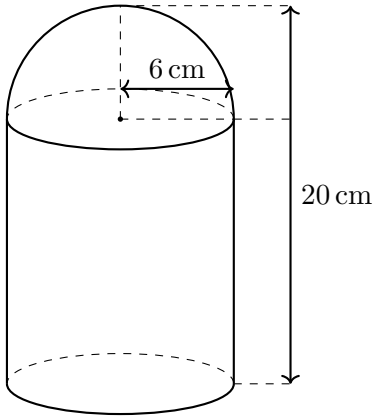
(a) Calculate the volume of the solid.

..... cm^3 **[3]**

(b) Calculate the total surface area of the solid.

..... cm^2 **[4]**

18 A solid is made by placing a solid hemisphere on top of a solid cylinder.
The cylinder and the hemisphere both have radius 6 cm.
The total height of the solid is 20 cm.



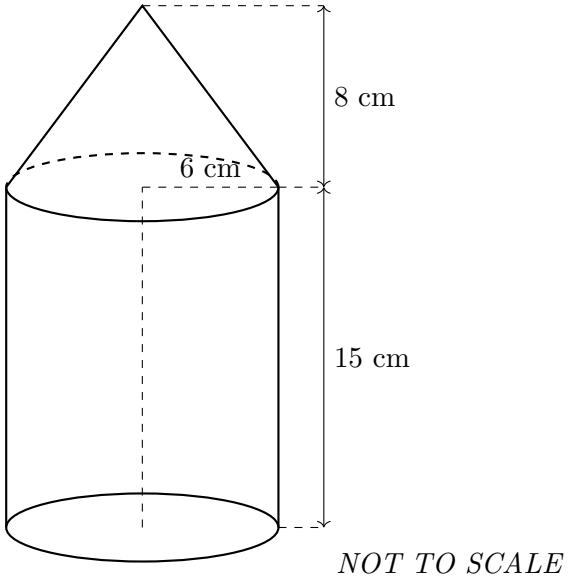
(a) Calculate the volume of the solid.

..... cm^3 **[3]**

(b) Calculate the total surface area of the solid.

..... cm^2 **[4]**

19 A solid is made from a cylinder and a cone.
The cylinder has radius 6 cm and height 15 cm.
The cone has radius 6 cm and height 8 cm.



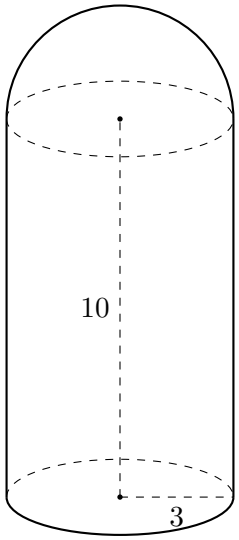
(a) Calculate the total volume of the solid.

Volume = ... cm^3 **[3]**

(b) Calculate the total surface area of the solid.

Surface area = ... cm^2 **[4]**

20 A solid is made by attaching a hemisphere of radius 3 cm to one end of a solid cylinder of radius 3 cm and height 10 cm.



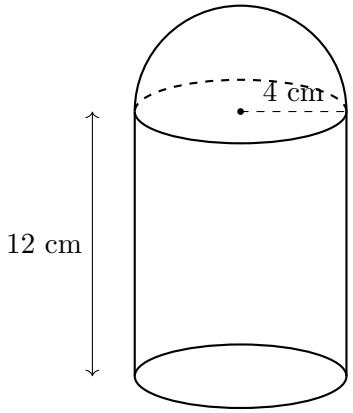
(a) Calculate the volume of the solid.

Volume = ... cm^3 **[3]**

(b) Calculate the total surface area of the solid.

Total surface area = ... cm^2 **[4]**

21 A solid object is formed by joining a hemisphere to one end of a solid cylinder.
The cylinder has a radius of 4 cm and a length of 12 cm.
The hemisphere has the same radius as the cylinder.



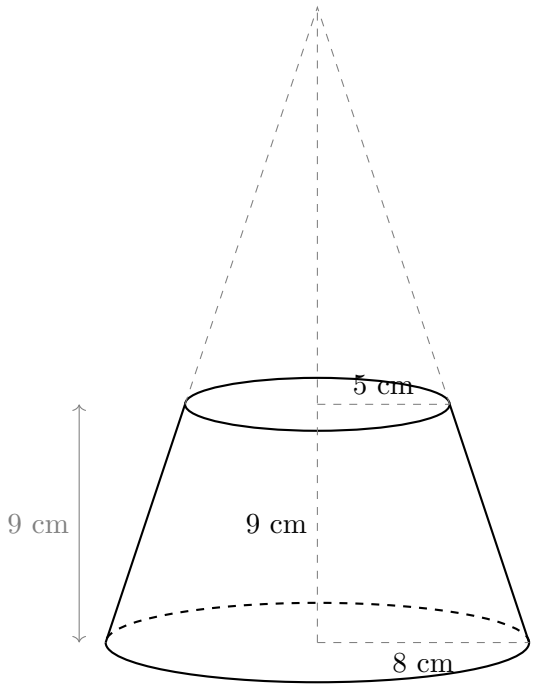
(a) Calculate the volume of the solid.

Volume = ... cm^3 **[3]**

(b) Calculate the total surface area of the solid.

Surface area = ... cm^2 **[4]**

22 The diagram shows a solid frustum.
The frustum is made by removing a smaller cone from a larger cone.
The larger cone has a base radius of 8 cm and a vertical height of 24 cm.
The smaller cone has a base radius of 5 cm and a vertical height of 15 cm.



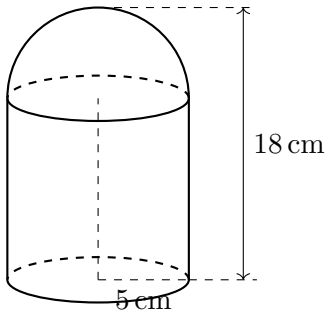
(a) Calculate the volume of the solid frustum.

..... cm^3 **[3]**

(b) Calculate the total surface area of the solid frustum.

..... cm^2 **[5]**

23 A solid is formed by placing a hemisphere of radius 5 cm on top of a cylinder of radius 5 cm.
The total height of the solid is 18 cm.



NOT TO SCALE

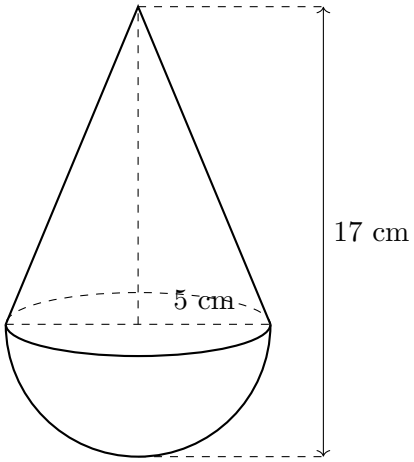
(a) Calculate the volume of the solid.
Give your answer correct to 3 significant figures.

..... cm^3 **[4]**

(b) Calculate the total surface area of the solid.
Give your answer correct to 3 significant figures.

..... cm^2 **[4]**

24 The diagram shows a solid made by joining a solid hemisphere to a solid cone.
The common radius of the hemisphere and the base of the cone is 5 cm.
The total height of the solid is 17 cm.



(a) Calculate the volume of the solid.

Volume = ... cm^3 **[3]**

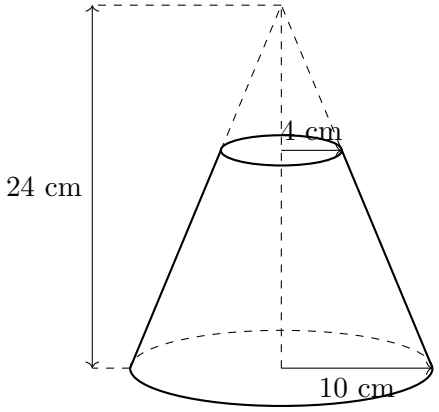
(b) Calculate the total surface area of the solid.

Surface area = ... cm^2 **[3]**

(c) The solid is melted down and completely recast into a single solid cylinder of radius 4 cm.
Calculate the height of the cylinder.

Height = ... cm **[3]**

25 A solid cone of height 24 cm and base radius 10 cm is cut parallel to its base to form a frustum and a smaller cone.
The radius of the top circular face of the frustum is 4 cm.



(a) Show that the height of the removed small cone is 9.6 cm.

[2]

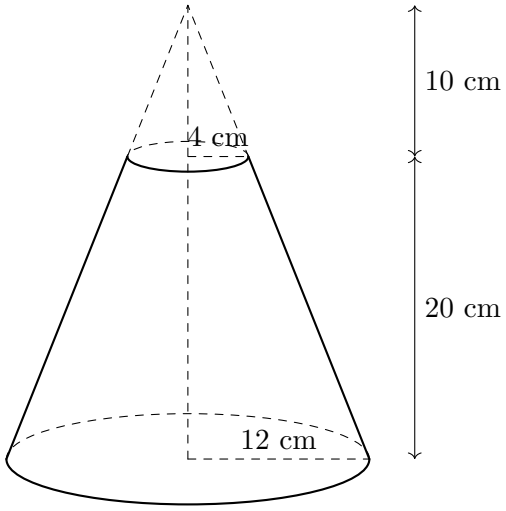
(b) Calculate the volume of the frustum.

..... cm^3 **[3]**

(c) The solid frustum is melted down and recast entirely into a solid sphere.
Calculate the radius of this sphere.

..... cm **[3]**

26 The diagram shows a solid metal frustum.
The frustum is formed by removing a small cone of height 10 cm from a larger cone.
The radius of the circular base of the frustum is 12 cm and the radius of the circular top is 4 cm.
The height of the frustum is 20 cm.



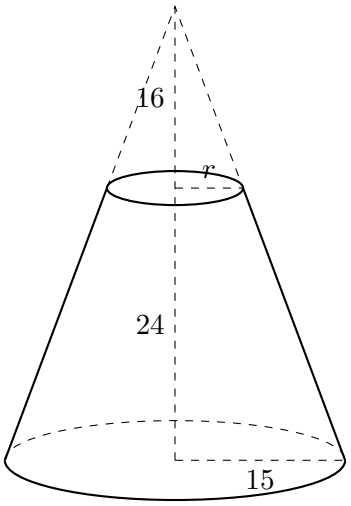
(a) Calculate the volume of the solid frustum.

..... cm^3 **[4]**

(b) Calculate the total surface area of the solid frustum.

..... cm^2 **[4]**

27 A container is in the shape of a frustum of a cone.
The original cone has a base radius of 15 cm and a vertical height of 40 cm.
A smaller cone of vertical height 16 cm is removed from the top to form the frustum.



(a) Show that the radius r of the top circular face of the frustum is 6 cm.

[2]

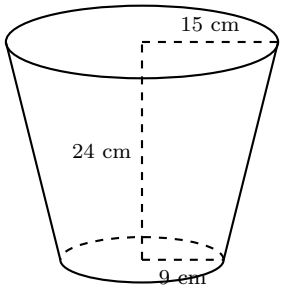
(b) Calculate the volume of the frustum.

Volume = ... cm^3 **[3]**

(c) The solid frustum is melted down and recast entirely into a solid sphere.
Calculate the radius of the sphere.

Radius = ... cm **[3]**

28 The diagram shows an open bucket in the shape of a frustum of a right circular cone.
The circular base has radius 9 cm and the open circular top has radius 15 cm.
The perpendicular height of the bucket is 24 cm.



(a) Show that the perpendicular height of the original cone before the bottom was removed is 60 cm.

[2]

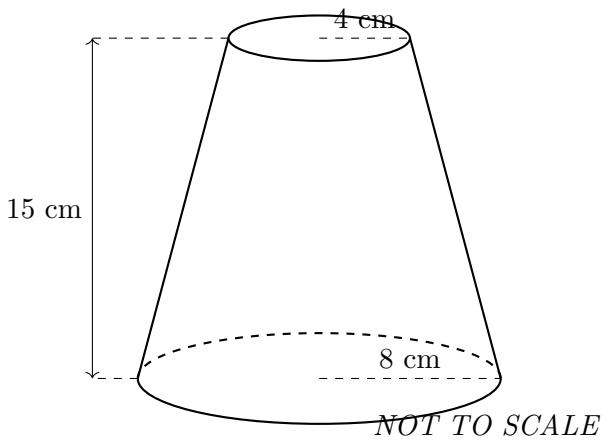
(b) Calculate the capacity of the bucket in litres.

Capacity = ... litres **[3]**

(c) Calculate the total external surface area of the open bucket.

Surface area = ... cm^2 **[3]**

29 A solid frustum of a cone has top radius 4 cm, bottom radius 8 cm and vertical height 15 cm.
It is formed by removing a smaller cone from the top of a larger cone.



(a) Show that the original height of the cone before the top was removed was 30 cm.

[2]

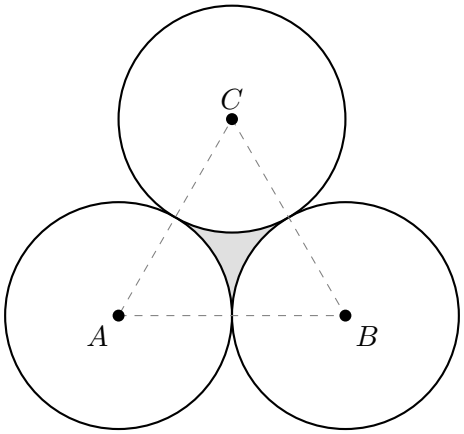
(b) Calculate the volume of the frustum.

Volume = ... cm^3 **[3]**

(c) Calculate the total surface area of the solid frustum.

Surface area = ... cm^2 **[4]**

30 The diagram shows three identical circles, each of radius r cm.
The circles touch each other externally.
The region enclosed between the three circles is shaded.



(a) Show that the area of the shaded region, A, is given by the formula

$$A = r^2\left(\sqrt{3} - \frac{\pi}{2}\right)$$

[4]

(b) The area of the shaded region is 15 cm^2.

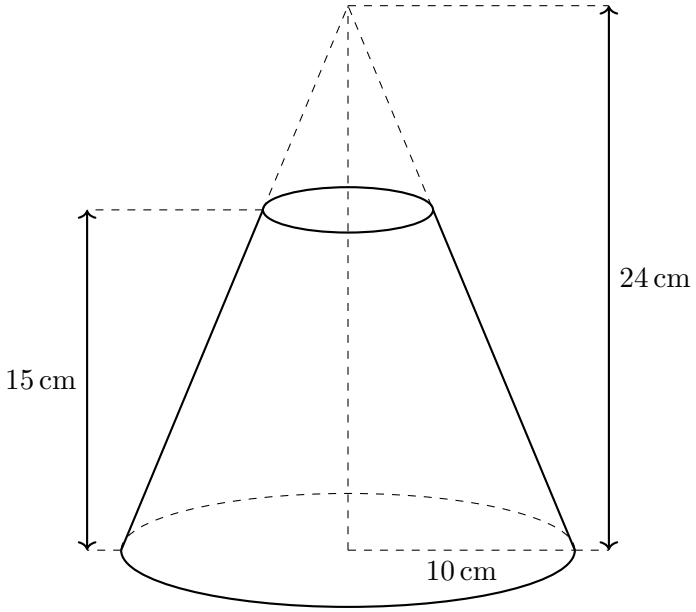
Work out the value of r.

$r =$ ..... **[3]**

(c) Calculate the total perimeter of the shaded region.

..... cm **[3]**

31 A frustum is made by removing a smaller cone from the top of a larger cone.
The larger cone has base radius 10 cm and vertical height 24 cm.
The frustum has vertical height 15 cm.



(a) Find the radius of the top of the frustum.

..... cm **[2]**

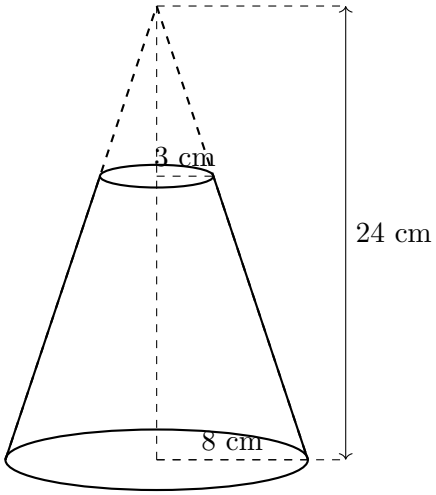
(b) Calculate the volume of the frustum.

..... cm^3 **[3]**

(c) Calculate the total surface area of the frustum.

..... cm^2 **[5]**

32 A frustum is made by removing a small cone from the top of a large cone.
The large cone has a base radius of 8 cm and a total height of 24 cm.
The removed small cone has a base radius of 3 cm.



(a) Show that the height of the small cone removed is 9 cm.

[2]

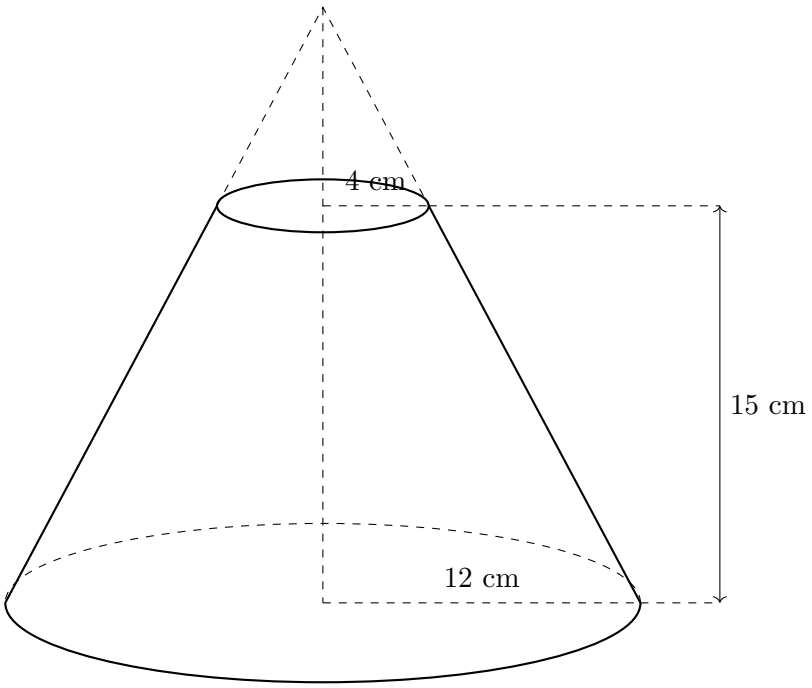
(b) Calculate the volume of the frustum.

Volume = ... cm^3 **[3]**

(c) Calculate the total surface area of the frustum.

Surface area = ... cm^2 **[5]**

33 The diagram shows a solid frustum of a cone.
The frustum is made by removing a small cone from the top of a larger solid cone.
The base radius of the frustum is 12 cm and the top radius is 4 cm.
The height of the frustum is 15 cm.



(a) Calculate the height of the small cone that was removed.

Height = ... cm **[3]**

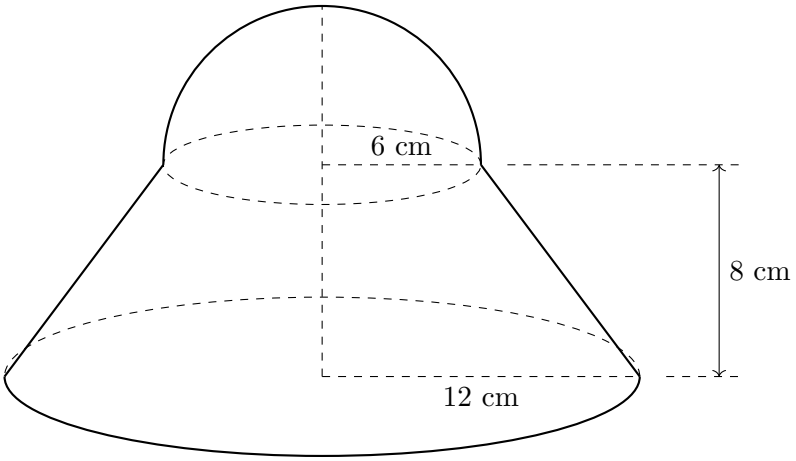
(b) Calculate the volume of the solid frustum.

Volume = ... cm^3 **[3]**

(c) Calculate the total surface area of the solid frustum.

Surface area = ... cm^2 **[5]**

34 The diagram shows a compound solid made from a frustum of a cone and a solid hemisphere.
The frustum is created by removing a small cone from a large cone.
The base radius of the frustum is 12 cm and its top radius is 6 cm.
The height of the frustum is 8 cm.
A solid hemisphere of radius 6 cm is attached to the top face of the frustum.



NOT TO SCALE

(a) Show that the perpendicular height of the large cone from which the frustum is made is 16 cm.

[3]

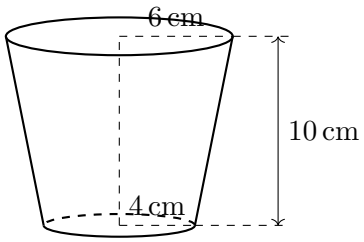
(b) Calculate the total volume of the compound solid.

..... cm^3 **[4]**

(c) Calculate the total surface area of the compound solid.

..... cm^2 **[4]**

35 The diagram shows a closed container in the shape of a frustum of a cone.
The radius of the top circular face is 6 cm and the radius of the base is 4 cm.
The vertical height of the container is 10 cm.



NOT TO SCALE

(a) The frustum is made by removing a smaller cone from a larger cone.
Calculate the vertical height of the larger cone.

..... cm **[3]**

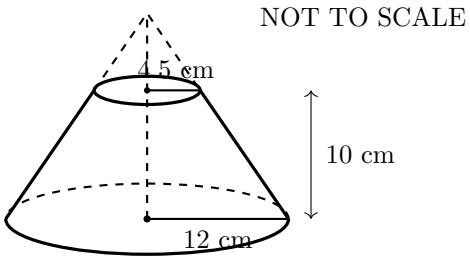
(b) Calculate the volume of the container.
Give your answer correct to 3 significant figures.

..... cm^3 **[3]**

(c) Calculate the total surface area of the closed container.
Give your answer correct to 3 significant figures.

..... cm^2 **[5]**

36 The diagram shows a solid frustum formed by removing a small right circular cone of base radius 4.5 cm from the top of a larger right circular cone of base radius 12 cm. The vertical height of the frustum is 10 cm.



(a) Use similar triangles to show that the vertical height of the removed cone is 6 cm.

[2]

(b) Calculate the total surface area of the solid frustum, giving your answer correct to 3 significant figures.

..... cm^2 **[4]**

(c) Calculate the volume of the solid frustum, giving your answer correct to 3 significant figures.

..... cm^3 **[3]**

(d) The solid frustum is melted down and recast into n identical solid spheres, each of radius 2.4 cm. Calculate the greatest possible integer value of n.

$n =$ ..... **[3]**

1 Identifies the three straight edges as $10 + 6 + 10 = 26$ **[M1]**

Calculates the arc length of the semicircle as $\frac{1}{2} \times \pi \times 6$ (or 3π) **[M1]**

$26 + 3\pi$ **[A1]**

[Back to Question] *[Total Marks: 3]*

2 (a) Arc length $BD = \frac{1}{4} \times 2 \times \pi \times 10 = 5\pi$ **[M1]**

Perimeter $= 10 + 10 + 5\pi = 20 + 5\pi$ **[A1]**

(b) Area of quarter circle $= \frac{1}{4} \times \pi \times 10^2 = 25\pi$ **[M1]**

Area of shaded region $= 100 - 25\pi$ **[A1]**

[Back to Question] *[Total Marks: 4]*

3 (a) Curved edge of semi-circle $= \frac{1}{2} \times \pi \times 4 = 2\pi$ **[M1]**

$20 + 2\pi$ **[A1]**

(b) $(8 \times 4) + \left(\frac{1}{2} \times \pi \times 2^2\right)$ **[M1]**

$32 + 2\pi$ **[A1]**

[Back to Question] *[Total Marks: 4]*

4 (a) Perimeter $= 12 + 12 + 8 + \frac{1}{2} \times \pi \times 8$ **[M1]**

$32 + 4\pi$ **[A1]**

(b) Area $= 12 \times 8 + \frac{1}{2} \times \pi \times 4^2$ **[M1]**

$96 + 8\pi$ **[A1]**

[Back to Question] *[Total Marks: 4]*

5 (a) Calculates semicircular arc length: $\frac{1}{2} \times \pi \times 14 = 7\pi$ **[M1]**

$50 + 7\pi$ **[A1]**

(b) Area of rectangle $= 18 \times 14 = 252$ **[B1]**

Area of semicircle $= \frac{1}{2} \times \pi \times 7^2 = 24.5\pi$ (or $\frac{49\pi}{2}$) **[M1]**

$252 + 24.5\pi$ (or $252 + \frac{49\pi}{2}$) **[A1]**

[Back to Question] *[Total Marks: 5]*

6 **(a)** Arc length $= \frac{1}{2} \times \pi \times 8 = 4\pi$ **[M1]**

$32 + 4\pi$ **[A1]**

(b) Area of rectangle $= 12 \times 8 = 96$ **[M1]**

Area of semicircle $= \frac{1}{2} \times \pi \times 4^2 = 8\pi$ **[M1]**

$96 + 8\pi$ **[A1]**

[Back to Question] *[Total Marks: 5]*

7 **(a)** Area of rectangle $= 10 \times 8 = 80$ **[M1]**

Area of semicircle $= \frac{1}{2} \times \pi \times 4^2 = 8\pi$ **[M1]**

$80 + 8\pi$ **[A1]**

(b) Identifies the three straight edges: $10 + 8 + 10 = 28$ **[M1]**

Perimeter of semicircle arc $= \frac{1}{2} \times \pi \times 8 = 4\pi$ **[M1]**

$28 + 4\pi$ **[A1]**

[Back to Question] *[Total Marks: 6]*

8 **(a)** Sum of three straight edges: $10 + 6 + 10 = 26$ **[M1]**

Arc length of semicircle $= \frac{1}{2} \times \pi \times 6$ $(= 3\pi)$ **[M1]**

$26 + 3\pi$ **[A1]**

(b) Area of rectangle $= 10 \times 6$ $(= 60)$ **[M1]**

Area of removed semicircle $= \frac{1}{2} \times \pi \times 3^2$ $(= 4.5\pi)$ **[M1]**

$60 - 4.5\pi$ **[A1]**

[Back to Question] *[Total Marks: 6]*

9 **(a)** Area of square $= 4 \times 4 = 16$ **[B1]**

Area of quarter circle $= \frac{1}{4} \times \pi \times 4^2 = 4\pi$ **[M1]**

$16 + 4\pi$ **[A1]**

(b) Arc length $= \frac{1}{4} \times 2 \times \pi \times 4 = 2\pi$ **[M1]**

Perimeter $= 4 + 4 + 4 + 2\pi$ **[M1]**

$12 + 2\pi$ **[A1]**

[Back to Question] *[Total Marks: 6]*

10 **(a)** Arc length $= \frac{1}{2} \times \pi \times 10$ $(= 5\pi)$ **[M1]**

Perimeter $= 12 + 12 + 10 + 5\pi$ **[M1]**

$34 + 5\pi$ **[A1]**

(b) Area of rectangle $= 12 \times 10$ $(= 120)$ **[M1]**

Area of semicircle $= \frac{1}{2} \times \pi \times 5^2$ $(= 12.5\pi)$ **[M1]**

$120 + 12.5\pi$ **[A1]**

[Back to Question] *[Total Marks: 6]*

11 **(a)** Calculates the area of the rectangle: $10 \times 6 = 60$ **[M1]**

Calculates the area of the semicircle: $\frac{1}{2} \times \pi \times 3^2$ $(= 4.5\pi)$ **[M1]**

$60 - 4.5\pi$ **[A1]**

(b) Calculates the sum of the three straight edges: $10 + 6 + 10 = 26$ **[M1]**

Calculates the arc length of the semicircle: $\frac{1}{2} \times \pi \times 6$ $(= 3\pi)$ **[M1]**

$26 + 3\pi$ **[A1]**

[Back to Question] *[Total Marks: 6]*

12 **(a)** **(i)** Semicircular arc length $= \frac{1}{2} \times 2\pi(6) = 6\pi$ cm **[M1]**

Outer perimeter $= 16 + 12 + 16 + 6\pi = 44 + 6\pi$ **[A1]**

(ii) Area of rectangle $= 16 \times 12 = 192\ cm^2$ **[B1]**

Area of semicircle $= \frac{1}{2}\pi(6^2) = 18\pi\ cm^2$ and area of hole $= \pi(4^2) = 16\pi\ cm^2$ **[M1]**

Shaded area $= 192 + 18\pi - 16\pi = 192 + 2\pi$ **[A1]**

(b) $5(192 + 2\pi)$ **[M1]**

$960 + 10\pi$ **[A1]**

[Back to Question] *[Total Marks: 7]*

13 **(a)** $h^2 = 13^2 - 5^2 = 169 - 25 = 144$ **[M1]**

$h = \sqrt{144} = 12$ cm **[A1]**

(b) Curved surface area of cone $= \pi(5)(13) = 65\pi\ cm^2$ **[B1]**

Curved surface area of cylinder $= 2\pi(5)(12) = 120\pi\ cm^2$ **[B1]**

Inner curved surface area of hemisphere $= \frac{1}{2} \times 4\pi(5^2) = 50\pi\ cm^2$ **[B1]**

Total surface area $= 65\pi + 120\pi + 50\pi = 235\pi$ **[A1]**

(c) Volume of cone $= \frac{1}{3}\pi(5^2)(12) = 100\pi$ and volume of cylinder $= \pi(5^2)(12) = 300\pi$ **[M1]**

Volume of hemisphere $= \frac{1}{2} \times \frac{4}{3}\pi(5^3) = \frac{250}{3}\pi$ **[B1]**

Total volume $= 100\pi + 300\pi - \frac{250}{3}\pi$ **[M1]**

$\frac{950}{3}\pi$ **[A1]**

[Back to Question] *[Total Marks: 10]*

14 **(a)** Volume of cylinder $= \pi \times 5^2 \times 12 = 300\pi$ **[M1]**

Volume of hemisphere $= \frac{2}{3} \times \pi \times 5^3 = \frac{250\pi}{3}$ **[M1]**

1200 (or 1204 or 1204.2 . . .) **[A1]**

(b) Curved surface of cylinder $= 2 \times \pi \times 5 \times 12 = 120\pi$ **[M1]**

Curved surface of hemisphere + Base of cylinder $= 2 \times \pi \times 5^2 + \pi \times 5^2 = 75\pi$ **[M1]**

613 (or 612.6 . . .) **[A1]**

[Back to Question] *[Total Marks: 6]*

15 **(a)** Volume of cylinder $= \pi \times 4^2 \times 10 = 160\pi$ $(= 502.65...)$ **[M1]**

Volume of cone $= \frac{1}{3} \times \pi \times 4^2 \times 7.5 = 40\pi$ $(= 125.66...)$ **[M1]**

628 (accept 628.3 or 200π) **[A1]**

(b) Finds slant height of cone: $l = \sqrt{4^2 + 7.5^2} = 8.5$ **[B1]**

Sums base area, curved cylinder area, and curved cone area:
$\pi(4)^2 + 2\pi(4)(10) + \pi(4)(8.5) = 130\pi$ **[M1]**

408 (accept 408.4 or 130π) **[A1]**

[Back to Question] *[Total Marks: 6]*

16 **(a)** Height of cylinder $= 25 - 5 = 20$ **[B1]**

Total Volume $= \pi \times 5^2 \times 20 + \frac{2}{3} \times \pi \times 5^3$ **[M1]**

1830 or 1832.5 . . . **[A1]**

(b) Curved surface area of cylinder $= 2 \times \pi \times 5 \times 20$ $(= 200\pi)$ **[M1]**

Area of circular base $= \pi \times 5^2$ $(= 25\pi)$ **[M1]**

Curved surface area of hemisphere $= 2 \times \pi \times 5^2$ $(= 50\pi)$ **[M1]**

864 or 863.9 . . . **[A1]**

[Back to Question] *[Total Marks: 7]*

17 **(a)** Volume of cylinder $= \pi \times 5^2 \times 12$ $(= 300\pi$ or 942.47...) **[M1]**

Volume of cone $= \frac{1}{3} \times \pi \times 5^2 \times 6$ $(= 50\pi$ or 157.07...) **[M1]**

1100 or 1099.5... to 1099.6 **[A1]**

(b) Slant height of cone $= \sqrt{5^2 + 6^2}$ $(= \sqrt{61}$ or 7.81...) **[M1]**

Curved surface area of cone $= \pi \times 5 \times$ their $\sqrt{61}$ $(= 122.68...)$ **[M1]**

Curved surface area of cylinder + base $= (2 \times \pi \times 5 \times 12) + (\pi \times 5^2)$ $(= 145\pi$ or 455.53...) **[M1]**

578 or 578.2... **[A1]**

[Back to Question] *[Total Marks: 7]*

18 (a) Volume of cylinder $= \pi \times 6^2 \times 14$ **[M1]**

Volume of hemisphere $= \frac{2}{3} \times \pi \times 6^3$ **[M1]**

2040 or 2035.7... **[A1]**

(b) Curved surface area of cylinder + base area $= 2 \times \pi \times 6 \times 14 + \pi \times 6^2$ (or $168\pi + 36\pi$) **[M1]**

Curved surface area of hemisphere $= 2 \times \pi \times 6^2$ (or 72π) **[M1]**

Correct addition of all three parts **[M1]**

867 or 867.0... to 867.1 **[A1]**

[Back to Question] *[Total Marks: 7]*

19 (a) Volume of cylinder $= \pi \times 6^2 \times 15 = 540\pi$ **[M1]**

Volume of cone $= \frac{1}{3} \times \pi \times 6^2 \times 8 = 96\pi$ **[M1]**

1998 or 2000 (to 3 sf) **[A1]**

(b) Slant height of cone $l = \sqrt{6^2 + 8^2} = 10$ cm **[M1]**

Curved surface area of cone $= \pi \times 6 \times 10 = 60\pi$ **[M1]**

Curved surface area of cylinder + base area $=$
$2 \times \pi \times 6 \times 15 + \pi \times 6^2 = 180\pi + 36\pi = 216\pi$ **[M1]**

867 (to 3 sf) **[A1]**

[Back to Question] *[Total Marks: 7]*

20 (a) Volume of cylinder $= \pi \times 3^2 \times 10 = 90\pi$ $(282.7...)$ **[M1]**

Volume of hemisphere $= \frac{2}{3} \times \pi \times 3^3 = 18\pi$ $(56.5...)$ **[M1]**

108π or 339 or 339.2 to 339.3 **[A1]**

(b) Area of circular base $= \pi \times 3^2 = 9\pi$ $(28.2...)$ **[B1]**

Curved area of cylinder $= 2 \times \pi \times 3 \times 10 = 60\pi$ $(188.4...)$ **[M1]**

Curved area of hemisphere $= 2 \times \pi \times 3^2 = 18\pi$ $(56.5...)$ **[M1]**

87π or 273 or 273.3 to 273.4 **[A1]**

[Back to Question] *[Total Marks: 7]*

21 **(a)** $\pi \times 4^2 \times 12$ (or 192π) **[M1]**

$\frac{2}{3} \times \pi \times 4^3$ (or $\frac{128\pi}{3}$) **[M1]**

737 or 737.2... or $\frac{704\pi}{3}$ **[A1]**

(b) Calculates curved surface area of cylinder: $2 \times \pi \times 4 \times 12$ (or 96π) **[M1]**

Calculates curved surface area of hemisphere: $2 \times \pi \times 4^2$ (or 32π) **[M1]**

Adds all three external faces including the flat circular base: $96\pi + 32\pi + \pi(4^2)$ **[M1]**

452 or 452.3... to 452.4 or 144π **[A1]**

[Back to Question] *[Total Marks: 7]*

22 **(a)** Volume of large cone $= \frac{1}{3} \times \pi \times 8^2 \times 24$ (=512π) **[M1]**

Volume of small cone $= \frac{1}{3} \times \pi \times 5^2 \times 15$ (=125π) **[M1]**

1215.8 (or 1220 or 387π) **[A1]**

(b) Slant height of large cone $= \sqrt{8^2 + 24^2}$ (=$8\sqrt{10}$ or 25.298...) **[M1]**

Slant height of small cone $= \sqrt{5^2 + 15^2}$ (=$5\sqrt{10}$ or 15.811...) **[M1]**

Curved SA of frustum $= \pi \times 8 \times 8\sqrt{10} - \pi \times 5 \times 5\sqrt{10}$ (=387.4...) **[M1]**

Total SA = Curved SA + $\pi(8^2) + \pi(5^2)$ **[M1]**

667 (or 667.0...) **[A1]**

[Back to Question] *[Total Marks: 8]*

23 **(a)** Calculates the height of the cylinder: $18 - 5 = 13$ cm **[B1]**

Calculates the volume of the cylinder: $\pi \times 5^2 \times 13$ (= 325π) **[M1]**

Calculates the volume of the hemisphere: $\frac{2}{3} \times \pi \times 5^3$ (= $\frac{250\pi}{3}$) **[M1]**

1280 (accept 1280 to 1284) **[A1]**

(b) Calculates the curved surface area of the cylinder: $2 \times \pi \times 5 \times 13$ (= 130π) **[M1]**

Calculates the curved surface area of the hemisphere: $2 \times \pi \times 5^2$ (= 50π) **[M1]**

Calculates the area of the circular base: $\pi \times 5^2$ (= 25π) **[M1]**

644 (accept 643 to 645) **[A1]**

[Back to Question] *[Total Marks: 8]*

24 **(a)** Height of cone = $17 - 5 = 12$ cm **[M1]**

Volume of cone $= \frac{1}{3} \times \pi \times 5^2 \times 12$ (100π or 314.1...) AND Volume of hemisphere = $\frac{2}{3} \times \pi \times 5^3$ ($\frac{250}{3}\pi$ or 261.7...) **[M1]**

576 (accept 575.9 to 576.0) **[A1]**

(b) Slant height of cone $l = \sqrt{5^2 + 12^2} = 13$ **[M1]**

Curved SA of cone $= \pi \times 5 \times 13$ (65π) AND Curved SA of hemisphere $= 2 \times \pi \times 5^2$ (50π) **[M1]**

361 (accept 361.2 to 361.3) **[A1]**

(c) $\pi \times 4^2 \times h = 575.9...$ (or $\frac{550}{3}\pi$) **[M1]**

$16h = 183.3...$ (or $\frac{550}{3}$) **[M1]**

11.5 (accept 11.45 to 11.46) **[A1]**

[Back to Question] *[Total Marks: 9]*

25 (a) Uses similar triangles: $\frac{h}{24} = \frac{4}{10}$ **[M1]**

$h = \frac{24 \times 4}{10} = 9.6$ cm **[A1]**

(b) Volume of large cone $= \frac{1}{3} \times \pi \times 10^2 \times 24 = 800\pi$ **[M1]**

Volume of small cone $= \frac{1}{3} \times \pi \times 4^2 \times 9.6 = 51.2\pi$ **[M1]**

2350 (or 2352.4...) **[A1]**

(c) Sets up equation: $\frac{4}{3} \times \pi \times r^3 = 748.8\pi$ (or 2352.4) **[M1]**

$r^3 = 561.6$ **[M1]**

8.25 (or 8.250...) **[A1]**

[Back to Question] *[Total Marks: 8]*

26 (a) Total height of large cone $= 20 + 10 = 30$ cm **[B1]**

Volume of large cone $= \frac{1}{3} \times \pi \times 12^2 \times 30$ $(= 1440\pi$ or 4523.89...) **[M1]**

Volume of small cone $= \frac{1}{3} \times \pi \times 4^2 \times 10$ $(= 53.33\pi$ or 167.55...) **[M1]**

4360 or 4356... **[A1]**

(b) Slant height of large cone, $L = \sqrt{12^2 + 30^2}$ $(= \sqrt{1044}$ or 32.31...) AND slant height of small cone, $l = \sqrt{4^2 + 10^2}$ $(= \sqrt{116}$ or 10.77...) **[M1]**

Curved surface area of frustum $= \pi \times 12 \times \sqrt{1044} - \pi \times 4 \times \sqrt{116}$ $(= 1082.7...)$ **[M1]**

Total surface area $= 1082.7... + \pi \times 12^2 + \pi \times 4^2$ **[M1]**

1585 or 1585.3... to 1585.4 **[A1]**

[Back to Question] *[Total Marks: 8]*

27 (a) Uses similar triangles: $\frac{r}{15} = \frac{16}{40}$ **[M1]**

$r = 15 \times \frac{16}{40} = 6$ leading correctly to given answer **[A1]**

(b) Volume of large cone $= \frac{1}{3} \times \pi \times 15^2 \times 40 = 3000\pi$ (9424.7...) **[M1]**

Volume of small cone $= \frac{1}{3} \times \pi \times 6^2 \times 16 = 192\pi$ (603.1...) **[M1]**

2808π or 8820 or 8821.5 to 8822 **[A1]**

(c) Equating volumes: $\frac{4}{3} \times \pi \times R^3 = 2808\pi$ **[M1]**

$R^3 = 2106$ **[M1]**

$R = \sqrt[3]{2106} = 12.8$ or 12.81 to 12.82 **[A1]**

[Back to Question] *[Total Marks: 8]*

28 **(a)** Uses similar triangles: $\frac{h-24}{h} = \frac{9}{15}$ (or $\frac{x}{x+24} = \frac{9}{15}$) **[M1]**

Solves equation to obtain $h = 60$ **[A1]**

(b) Volume of large cone $= \frac{1}{3}\pi(15)^2(60) = 4500\pi$ and volume of small cone $= \frac{1}{3}\pi(9)^2(36) = 972\pi$ **[M1]**

Volume of frustum $= 4500\pi - 972\pi = 3528\pi = 11\,083.5...$ cm^3 **[M1]**

11.1 (accept 11.08) **[A1]**

(c) Finds slant heights: $L = \sqrt{15^2 + 60^2} = 61.846...$ and $l = \sqrt{9^2 + 36^2} = 37.107...$ (or $l_{frustum} = \sqrt{6^2 + 24^2} = 24.738...$) **[M1]**

Calculates curved surface area plus circular base: $\pi(15)(61.846...) - \pi(9)(37.107...) + \pi(9)^2$ **[M1]**

2120 (accept 2120 to 2121) **[A1]**

[Back to Question] *[Total Marks: 8]*

29 **(a)** Let x be the height of the removed cone. $\frac{x}{4} = \frac{x+15}{8}$ **[M1]**

$8x = 4x + 60 \implies 4x = 60 \implies x = 15$. Total height $= 15 + 15 = 30$ cm. **[A1]**

(b) Volume of large cone $= \frac{1}{3} \times \pi \times 8^2 \times 30 = 640\pi$ **[M1]**

Volume of small cone $= \frac{1}{3} \times \pi \times 4^2 \times 15 = 80\pi$ **[M1]**

1760 (to 3 sf) **[A1]**

(c) Slant height of large cone $L = \sqrt{8^2 + 30^2} \approx 31.05$ cm and small cone $l = \sqrt{4^2 + 15^2} \approx 15.52$ cm **[M1]**

Curved surface area of frustum $= \pi \times 8 \times \sqrt{964} - \pi \times 4 \times \sqrt{241} \approx 585.25$ **[M1]**

Area of top and bottom bases $= \pi \times 4^2 + \pi \times 8^2 = 80\pi \approx 251.33$ **[M1]**

837 (to 3 sf) **[A1]**

[Back to Question] *[Total Marks: 9]*

30 **(a)** Identifies that the centers form an equilateral triangle of side $2r$ **[M1]**

Area of triangle $= \frac{1}{2} \times 2r \times 2r \times \sin(60°) = r^2\sqrt{3}$ **[M1]**

Identifies 3 sectors inside the triangle, each with angle 60°: Total sector area $= 3 \times \frac{60}{360} \times \pi r^2 = \frac{\pi}{2}r^2$ **[M1]**

Area $= r^2\sqrt{3} - \frac{\pi}{2}r^2 = r^2\left(\sqrt{3} - \frac{\pi}{2}\right)$ with no errors seen **[A1]**

(b) $r^2\left(\sqrt{3} - \frac{\pi}{2}\right) = 15$ **[M1]**

$r^2 = \frac{15}{\sqrt{3} - \pi/2}$ (≈ 93.02) **[M1]**

9.64 (or 9.644...) **[A1]**

(c) Perimeter consists of 3 arcs, each with angle 60° **[M1]**

Total perimeter $= 3 \times \frac{60}{360} \times 2\pi(9.644\ldots) = \pi(9.644\ldots)$ **[M1]**

30.3 (or 30.29 to 30.30) **[A1]**

[Back to Question] *[Total Marks: 10]*

31 **(a)** $\frac{r}{10} = \frac{24 - 15}{24}$ or $\frac{r}{10} = \frac{9}{24}$ **[M1]**

3.75 **[A1]**

(b) Volume of large cone $= \frac{1}{3} \times \pi \times 10^2 \times 24$ (or 800π) **[M1]**

Volume of small cone $= \frac{1}{3} \times \pi \times 3.75^2 \times 9$ (or 42.1875π) **[M1]**

2380 or 2381 (accept 2380.7...) **[A1]**

(c) Slant height of large cone $L = \sqrt{10^2 + 24^2} = 26$ **[M1]**

Slant height of small cone $l = \sqrt{3.75^2 + 9^2} = 9.75$ **[M1]**

Curved surface area of frustum $= \pi \times 10 \times 26 - \pi \times 3.75 \times 9.75$ **[M1]**

Adding base and top areas: $+\pi \times 10^2 + \pi \times 3.75^2$ **[M1]**

1060 or 1060.2... to 1060.3 **[A1]**

[Back to Question] *[Total Marks: 10]*

32 **(a)** Uses similar triangles: $\frac{h}{24} = \frac{3}{8}$ **[M1]**

$h = \frac{3 \times 24}{8} = 9$ **[A1]**

(b) Volume of large cone $= \frac{1}{3} \times \pi \times 8^2 \times 24$ (or 512π) **[M1]**

Volume of small cone $= \frac{1}{3} \times \pi \times 3^2 \times 9$ (or 27π) **[M1]**

1520 or 1524 or 485π **[A1]**

(c) Calculates slant height of large cone $L = \sqrt{8^2 + 24^2}$ (or $\sqrt{640}$) and small cone $l = \sqrt{3^2 + 9^2}$ (or $\sqrt{90}$) **[M1]**

Curved surface area of frustum $= \pi(8)(\sqrt{640}) - \pi(3)(\sqrt{90})$ **[M1]**

Calculates area of both flat circular bases: $\pi(8^2) + \pi(3^2)$ (or 73π) **[M1]**

Adds curved surface area and both base areas together **[M1]**

776 (or 775.5... to 775.6) **[A1]**

[Back to Question] *[Total Marks: 10]*

33 (a) Let h be the height of the small cone. $\frac{h}{4} = \frac{h+15}{12}$ **[M1]**

$12h = 4h + 60$ **[M1]**

$h = 7.5$ **[A1]**

(b) Volume of large cone $= \frac{1}{3} \times \pi \times 12^2 \times 22.5$ (1080π or 3392.9...) **[M1]**

Volume of small cone $= \frac{1}{3} \times \pi \times 4^2 \times 7.5$ (40π or 125.6...) **[M1]**

3270 (accept 3267 to 3268) **[A1]**

(c) Slant height of large cone $= \sqrt{12^2 + 22.5^2} = 25.5$ **[M1]**

Slant height of small cone $= \sqrt{4^2 + 7.5^2} = 8.5$ **[M1]**

Curved surface area of frustum $= \pi \times 12 \times 25.5 - \pi \times 4 \times 8.5$ ($306\pi - 34\pi = 272\pi$) **[M1]**

Area of both circular bases $= \pi \times 12^2 + \pi \times 4^2$ ($144\pi + 16\pi = 160\pi$) **[M1]**

1360 (accept 1357 to 1358) **[A1]**

[Back to Question] *[Total Marks: 11]*

34 (a) Let H be the height of the large cone. Height of small removed cone $= H - 8$ **[M1]**

Uses similar triangles: $\frac{H}{12} = \frac{H-8}{6}$ (or equivalent ratio) **[M1]**

$6H = 12H - 96$

6H = 96

H = $16 with no errors seen [A1]$

(b) Volume of frustum $= \frac{1}{3} \times \pi \times 12^2 \times 16 - \frac{1}{3} \times \pi \times 6^2 \times 8$ ($= 768\pi - 96\pi = 672\pi$)

[M2, M1 for one correct cone volume]

Volume of hemisphere $= \frac{2}{3} \times \pi \times 6^3$ ($= 144\pi$) **[M1]**

Total volume $= 672\pi + 144\pi = 816\pi = 2560$ or $2563.5\ldots$ **[A1]**

(c) Slant height of large cone $= \sqrt{12^2 + 16^2} = 20$ AND Slant height of small cone $= \sqrt{6^2 + 8^2} = 10$ **[M1]**

Curved surface area of frustum $= \pi \times 12 \times 20 - \pi \times 6 \times 10$ $(= 180\pi)$ **[M1]**

Total surface area $= 180\pi$ (frustum curved) $+\pi \times 12^2$ (base) $+2 \times \pi \times 6^2$ (hemisphere) **[M1]**

1240 or 1244 to 1244.2 **[A1]**

[Back to Question] *[Total Marks: 11]*

35 **(a)** Sets up an equation using similar triangles (e.g. let x be the height of the smaller cone): $\frac{x}{4} = \frac{x+10}{6}$ (or equivalent) **[M1]**

Solves for the height of the smaller cone: $6x = 4x + 40 \implies 2x = 40 \implies x = 20$ **[M1]**

Calculates the height of the larger cone: $20 + 10 = 30$ **[A1]**

(b) Calculates the volume of the larger cone: $\frac{1}{3} \times \pi \times 6^2 \times 30$ $(= 360\pi)$ **[M1]**

Calculates the volume of the smaller cone: $\frac{1}{3} \times \pi \times 4^2 \times 20$ $(= \frac{320\pi}{3})$ **[M1]**

796 (accept 795 to 797) **[A1]**

(c) Calculates the slant height of the larger cone: $\sqrt{6^2 + 30^2} = \sqrt{936}$ $(= 30.59...)$ **[M1]**

Calculates the slant height of the smaller cone: $\sqrt{4^2 + 20^2} = \sqrt{416}$ $(= 20.39...)$ **[M1]**

Calculates the curved surface area of the frustum: $\pi \times 6 \times \sqrt{936} - \pi \times 4 \times \sqrt{416}$ $(= 320.38...)$ **[M1]**

Calculates the sum of the areas of the circular top and base: $\pi \times 6^2 + \pi \times 4^2$ $(= 52\pi$ or 163.36...) **[M1]**

484 (accept 483.7 to 484.5) **[A1]**

[Back to Question] *[Total Marks: 11]*

36 **(a)** $\frac{h}{h+10} = \frac{4.5}{12}$ $(= \frac{3}{8})$ **[M1]**

$8h = 3h + 30$ giving $5h = 30$ and $h = 6$ cm **[A1]**

(b) Slant height of full cone $= \sqrt{12^2 + 16^2} = 20$ cm and slant height of removed cone $= \sqrt{4.5^2 + 6^2} = 7.5$ cm **[M1]**

Curved surface area of frustum $= \pi(12)(20) - \pi(4.5)(7.5) = 206.25\pi$ cm^2 **[M1]**

Total surface area $= 206.25\pi + \pi(12^2) + \pi(4.5^2) = 370.5\pi$ $(= 1163.96\ldots)$ **[M1]**

1160 **[A1]**

(c) Volume $= \frac{1}{3}\pi(12^2)(16) - \frac{1}{3}\pi(4.5^2)(6)$ **[M1]**

$768\pi - 40.5\pi = 727.5\pi$ $(= 2285.508\ldots)$ **[M1]**

2290 **[A1]**

(d) Volume of one sphere $= \frac{4}{3}\pi(2.4^3) = 18.432\pi$ $(= 57.9058\ldots)$ cm^3 **[M1]**

$n = \frac{727.5\pi}{18.432\pi} = 39.469\ldots$ **[M1]**

39 **[A1]**

[Back to Question] *[Total Marks: 12]*